

Bluestock Mutual Fund

Analytics Capstone Project

Final Report — All 6 Days

Prepared by: Vatsal Awasthi

Bluestock Fintech Internship | June–July 2026

Repository: github.com/vatsalawasthi/CapstoneProject-Vatsal-

6 Days Completed	40 Funds Analysed	46K NAV Records	32.8K Transactions	11 DB Tables	15+ Charts
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Project Overview

This report documents the complete end-to-end data analytics pipeline built for the Bluestock Mutual Fund capstone project. Over 6 days, the project progressed from raw CSV ingestion through data cleaning, exploratory analysis, quantitative performance metrics, interactive dashboard development, and advanced risk analytics — all built on real mutual fund data covering 40 schemes across 10 fund houses.

Technology Stack

- Python 3.14 — pandas, numpy, scipy, matplotlib, seaborn, plotly, sqlalchemy
- SQLite — star schema database with 11 tables loaded via SQLAlchemy
- Jupyter Notebooks — 5 analytical notebooks with executed outputs
- Power BI Desktop — 4-page interactive dashboard
- GitHub — full version control, 7 commits across 6 days

Dataset Summary

- 10 raw CSV files covering 40 mutual fund schemes across 10 AMCs
- NAV history: January 2022 – May 2026 (~4.4 years, 46,000 trading-day rows)
- 32,778 investor transactions across 5,000 unique investors
- 7 benchmark indices: NIFTY50, NIFTY100, NIFTY_MIDCAP150, BSE_SMALLCAP, NIFTY500, CRISIL_LIQUID, CRISIL_GILT
- Portfolio holdings for 34 equity funds (6 debt funds excluded as expected)

Day 1 — Project Setup + Data Ingestion (ETL)

Due: 24 Jun 2026

Status: DONE

Time Estimate: 6–8 hours

Objectives

- Create project folder structure and initialise Git repository
- Install all dependencies and generate requirements.txt
- Load all 10 CSV datasets using pandas and profile each one
- Fetch live NAV data from mfapi.in for 6 key schemes
- Validate AMFI scheme code consistency across files

What Was Built

scripts/etl_pipeline.py

- Loads every CSV in data/raw/ using pandas
- Prints .shape, .dtypes, and .head() for each file
- Flags anomalies: null columns, duplicate rows, fully-empty columns
- Auto-detects fund_master and nav_history files by filename
- Cross-validates every AMFI/scheme code between fund_master and nav_history
- Writes full data quality summary to reports/

scripts/live_nav_fetch.py

- Hits GET https://api.mfapi.in/mf/{scheme_code} for 6 schemes
- Parses JSON response (meta + historical NAV data)
- Saves each scheme's NAV history as raw CSV in data/raw/
- Flags mismatches between task-brief labels and actual API scheme names

Key Findings

- All 10 CSVs loaded successfully with no encoding or parse errors
- 02_nav_history.csv: 40 funds × 1,150 trading days = 46,000 rows, all weekday dates, zero weekend rows
- 01_fund_master.csv: 40 schemes, 10 fund houses, 2 categories (Equity/Debt), 15 sub-categories, 5 risk grades

- All AMFI codes in fund_master matched nav_history — no orphaned codes in either direction
- API mismatch discovered: scheme code 125497 returned 'SBI Small Cap Fund', not 'HDFC Top 100' as labelled in task brief — real-world data quality example

Deliverables

File / Output	Description
scripts/etl_pipeline.py	Full CSV profiling, anomaly detection, AMFI code validation
scripts/live_nav_fetch.py	Live NAV fetch from mfapi.in for 6 scheme codes
requirements.txt	All project dependencies pinned
notebooks/01_data_ingestion.ipynb	Interactive notebook with profiling outputs
GitHub commit	Day 1: Data ingestion complete

Day 2 — Data Cleaning + SQL Database Design

Due: 25 Jun 2026

Status: DONE

Time Estimate: 7–8 hours

Objectives

- Clean the three primary datasets with bespoke rules per the task brief
- Run basic validation and standardisation on the remaining 7 supporting CSVs
- Design and implement a SQLite star schema with 11 tables
- Load all cleaned datasets into SQLite via SQLAlchemy and verify row counts
- Write 10 analytical SQL queries covering all key business questions
- Create a comprehensive data dictionary

Data Cleaning — Primary Files

02_nav_history.csv

- Parsed dates to datetime; sorted by amfi_code + date
- Validated NAV > 0: all 46,000 rows passed
- Built complete calendar per fund and forward-filled NAV for weekends/holidays
- Result: 46,000 trading-day rows expanded to 64,320 calendar-day rows
- 18,320 forward-filled rows clearly flagged with is_filled = True

08_investor_transactions.csv

- Standardised transaction_type to exactly: SIP / Lumpsum / Redemption
- Validated amount_inr > 0: all 32,778 rows passed
- Parsed transaction_date to datetime; validated kyc_status enum {Verified, Pending}
- Zero duplicates found — data was already clean

07_scheme_performance.csv

- Validated all return columns (1yr, 3yr, 5yr, benchmark_3yr) are numeric: all passed
- Expense ratio range check (0.1%–2.5%): all 40 funds within range (0.55%–1.64%)
- 3 of 40 rows flagged for sharpe_ratio_extreme_outlier: ICICI Pru Liquid, Kotak Liquid, ABSL Liquid
- Flagged rows kept — liquid fund Sharpe outlier is expected behavior, not a data error

SQLite Star Schema — 11 Tables

- dim_fund (40 rows) — one row per scheme, from fund_master
- dim_date (1,612 rows) — calendar dimension, generated in Python, 2022-01-01 to 2026-05-31
- fact_nav (64,320 rows) — daily NAV per fund. FK: amfi_code, date_key
- fact_transactions (32,778 rows) — investor transactions. FK: amfi_code, date_key
- fact_performance (40 rows) — point-in-time performance snapshot. FK: amfi_code
- fact_aum (90 rows) — AUM by fund house over time. FK: date_key
- 5 supporting reference tables: monthly_sip_inflows, category_inflows, industry_folio_count, portfolio_holdings, benchmark_indices

10 Analytical SQL Queries — Key Results

- Q1: Top AUM fund — Mirae Asset Emerging Bluechip at ₹49,046 Cr
- Q2: Average NAV grew from 207 (Jan 2022) to 357 (May 2026)
- Q3: SIP inflows grew from ₹11,517 Cr (2022) to ₹31,002 Cr (2025)
- Q4: Punjab leads by transaction count (2,965 transactions)
- Q5: 14 funds have expense_ratio < 1%
- Q6: SBI Small Cap leads by 3yr return at 23.39%
- Q7: SIP dominates by count (60%) but Lumpsum dominates by value (₹2.5L avg ticket)
- Q8: Banking sector dominates holdings at ₹62,840 Cr (19.3%)
- Q9: Small Cap attracted higher net inflows than Large Cap in FY25
- Q10: Net Banking has highest KYC-pending transaction volume (689 txns)

Deliverables

File / Output	Description
data/processed/*.csv	All 10 cleaned CSV files
data/db/bluestock_mf.db	Populated SQLite database (11 tables, ~107K rows)
sql/schema.sql	Full star schema DDL with PKs and FKs
sql/queries.sql	10 annotated analytical SQL queries
data_dictionary.md	Full column-level documentation for all 11 tables
notebooks/02_data_cleaning.ipynb	Interactive notebook with cleaning outputs
GitHub commit	Day 2: Cleaned data + SQLite DB loaded

Day 3 — Exploratory Data Analysis (EDA)

Due: 28 Jun 2026
Status: DONE
Time Estimate: 7–8 hours

Charts Produced (15+)

- Chart 1 — NAV Trend (Plotly): All 40 schemes 2022–2026, YoY growth annotated per year
- Chart 2 — AUM Growth (Seaborn): Grouped bar by fund house 2022–2025, SBI dominance highlighted
- Chart 3 — SIP Inflow (Plotly): Monthly trend Jan 2022–Dec 2025, all-time high annotated
- Chart 4 — Category Heatmap (Seaborn): 12-month × 12-category net inflow heatmap
- Chart 5 — Demographics: Age group pie, SIP amount box plot, gender split pie
- Chart 6 — Geographic: SIP amount by state horizontal bar, T30 vs B30 pie
- Chart 7 — Folio Growth: Line chart 13.26 Cr → 26.12 Cr, milestones annotated
- Chart 8 — Correlation Matrix (Seaborn): 10-fund pairwise daily return heatmap
- Chart 9 — Sector Donut (Plotly): Aggregate sector weights across all equity funds

10 Key EDA Findings

- Finding 1: NAV growth is smooth and steady (~12-14%/year) — no 2024 correction visible in actual data
- Finding 2: SBI Mutual Fund leads at ₹12.5L Cr by 2025 — nearly double its 2022 AUM
- Finding 3: SIP inflows grew 2.7× in 4 years, peaking at ₹31,002 Cr (Dec 2025)
- Finding 4: Category inflow data only covers 12 months — Small Cap > Large Cap in FY25
- Finding 5: 26–35 age group is the largest investor cohort (40.7%); 66.7% male
- Finding 6: T30 cities account for 66.7% of investors and 2× the SIP value of B30 cities
- Finding 7: Folio count doubled from 13.26 Cr to 26.12 Cr — crossed 20 Cr in Oct 2024
- Finding 8: NAV return correlations near zero across all funds — data generation characteristic
- Finding 9: Banking is the largest equity sector exposure at ₹62,840 Cr (19.3% of holdings)
- Finding 10: SIP dominates by transaction count (60%); Lumpsum by value (₹2.5L avg ticket)

Deliverables

File / Output	Description
notebooks/03_eda_analysis.ipynb	Executed notebook with all 9 chart sections and 10 findings

reports/charts/01-09_*.png	9 exported PNG charts for report and dashboard
GitHub commit	Day 3: EDA notebook complete — 15+ charts, 10 documented insights

Day 4 — Fund Performance Analytics

Due: 29 Jun 2026

Status: DONE

Time Estimate: 7–8 hours

Metrics Computed — All 40 Funds

Daily Returns

- Formula: $\text{daily_return} = \text{nav_t} / \text{nav_t-1} - 1$
- 45,960 observations across 40 funds (using trading-day-only NAV rows)
- Distribution: Laplace-like, centred at zero, fat tails from small-cap funds
- KEY: Forward-filled weekend rows excluded — including them understates volatility by ~15%

CAGR (Compound Annual Growth Rate)

- Formula: $\text{CAGR} = (\text{NAV_end} / \text{NAV_start})^{(1/n)} - 1$
- 1yr and 3yr computed fresh from NAV history (both within available 4.4yr range)
- 5yr: used pre-reported return_5yr_pct from scheme_performance.csv (NAV history insufficient)
- 1yr range: -42.8% to +82.9% | 3yr top fund: Axis Midcap at 35.1%

Sharpe and Sortino Ratios

- Sharpe = $(R_p - R_f) / \text{Std}(R_p) \times \sqrt{252}$. Risk-free rate R_f = 6.5% (RBI repo rate proxy)
- Sortino = same formula but denominator uses only downside standard deviation
- Several funds show negative Sharpe — mean return below 6.5% risk-free hurdle

Alpha & Beta (OLS vs Nifty 100)

- Method: `scipy.stats.linregress(nifty100_returns, fund_returns)`
- Mean R^2 across all 40 funds: < 1% — NAV series generated independently of market
- Beta values cluster near zero vs expected 0.8–1.2 for real equity funds

Maximum Drawdown

- Formula: $\min(\text{NAV} / \text{running_max} - 1)$ for each fund
- Range: -0.1% (Liquid funds) to -52.6% (Small Cap equity)
- Peak and trough dates recorded for each fund's worst drawdown period

Fund Scorecard (0–100 Composite)

- $30\% \times 3\text{yr return rank} + 25\% \times \text{Sharpe rank} + 20\% \times \text{Alpha rank}$
- $15\% \times \text{expense ratio rank (inverse)} + 10\% \times \text{max drawdown rank (inverse)}$
- All components converted to 0–100 percentile rank before weighting
- Score range: 12–86 across 40 funds — sensible spread with no ties

Deliverables

File / Output	Description
notebooks/04_performance_analytics.ipynb	Executed notebook with all metrics
fund_scorecard.csv	0–100 composite scorecard for all 40 funds
alpha_beta.csv	Alpha, Beta, and R^2 for all 40 funds vs Nifty 100
reports/charts/10_daily_return_distribution.png	Pooled daily return histogram + KDE
reports/charts/11_benchmark_comparison_top5.png	Top 5 funds vs Nifty 50/100
GitHub commit	Day 4: Fund Performance Analytics complete

Day 5 — Dashboard Development (Power BI)

Due: 30 Jun 2026

Status: DONE

Time Estimate: 7–8 hours

Dashboard Pages Built

Page 1 — Industry Overview

- KPI cards: Total AUM, SIP Inflows (₹31K Cr), Folios (26.12 Cr), Total Schemes
- Industry AUM trend line chart (2022–2025)
- AUM by Fund House horizontal bar chart

Page 2 — Fund Performance

- Scatter plot: 3yr return (X) vs Sharpe ratio (Y), bubble size = AUM across 40 funds
- Fund scorecard table (sortable by any column)
- NAV vs Benchmark line chart with scheme slicer
- Slicers: fund house, category, plan

Page 3 — Investor Analytics

- Avg SIP amount by age group bar chart
- Transaction type donut: SIP 60.15%, Lumpsum 24.7%, Redemption 15.15%
- SIP amount by state horizontal bar
- Monthly transaction volume line chart
- Slicers: state, age group, city tier

Page 4 — SIP & Market Trends

- Dual-axis: SIP inflow (bar) + Nifty 50 (line) 2022–2025
- Category net inflows matrix heatmap (Apr 2024 – Mar 2025)
- Top 5 categories by net inflow FY25

Data Model Note

The task brief's stated KPIs for Total AUM (₹81L Cr) and Schemes (1,908) did not match the actual cleaned data (₹62.74L Cr and 1,522 respectively). The dashboard displays the real values from the cleaned CSVs — SIP Inflows (₹31K Cr) and Folios (26.12 Cr) matched exactly.

Deliverables

File / Output	Description
dashboard/bluestock_mf_dashboard.pbix	4-page Power BI dashboard file
dashboard/Dashboard.pdf	PDF export of all 4 pages
dashboard/Page1-4_*.png	4 page PNG screenshots
GitHub commit	Day 5: Power BI dashboard complete

Day 6 — Advanced Analytics + Risk Metrics

Due: 02 Jul 2026

Status: DONE

Time Estimate: 6–7 hours

Advanced Metrics Built

Historical VaR (95%) and CVaR — All 40 Funds

- $\text{VaR}(95\%) = 5\text{th percentile of daily return distribution (historical simulation method)}$
- $\text{CVaR}(95\%) = \text{mean of all returns below the VaR threshold (Expected Shortfall)}$
- Small Cap funds: worst VaR (most negative, -3.5%+ daily loss at 5th percentile)
- Liquid/Gilt funds: best VaR (near 0%), consistent with their low-volatility nature

Rolling 90-Day Sharpe Ratio — 5 Key Funds

- Formula: $\text{returns.rolling}(90).\text{mean}() / \text{returns.rolling}(90).\text{std}() \times \sqrt{252}$
- 5 funds: SBI Bluechip, Axis Midcap, SBI Small Cap, ICICI Pru Liquid, SBI Magnum Gilt
- Rolling Sharpe is highly volatile — frequently swings positive/negative within the year

Investor Cohort Analysis

- Grouped investors by first transaction year (2024 vs 2025)
- 2024 cohort: 4,624 investors, top fund ICICI Pru Bluechip Direct
- 2025 cohort: 138 investors, top fund SBI Small Cap Direct — shift toward higher risk

SIP Continuity Analysis

- Investors with 6+ SIP transactions: 1,362 eligible
- At-risk (avg gap > 35 days): 1,332 investors (97.8%)
- High rate reflects non-monthly transaction date distribution in dataset

Simple Fund Recommender

- Input: risk appetite (Low / Moderate / High)
- Logic: filter by risk_category, rank by Sharpe ratio, return top 3 funds
- Low → Liquid/Gilt funds | Moderate → Large Cap | High → Mid/Small Cap
- Saved as standalone scripts/recommender.py callable from command line

Sector HHI Concentration

- Herfindahl-Hirschman Index = $\sum(\text{weight}_i^2) \times 10,000$ per fund
- 73.5% of equity funds: Low concentration ($\text{HHI} < 1,500$) — well diversified
- 26.5%: Moderate concentration ($1,500 \leq \text{HHI} < 2,500$)
- No fund in High concentration range ($\text{HHI} \geq 2,500$)

Deliverables

File / Output	Description
notebooks/05_advanced_analytics.ipynb	Executed notebook with all 6 analyses
var_cvar_report.csv	VaR and CVaR for all 40 funds
scripts/recommender.py	Standalone fund recommender script
reports/charts/12-16_*.png	5 advanced analytics charts
reports/at_risk_sip_investors.csv	List of at-risk SIP investors
GitHub commit	Day 6: Advanced Analytics complete

Project Summary & Key Insights

6	5	11	10	15+	7
Days Done	Notebooks	DB Tables	SQL Queries	Charts	Git Commits

Top 10 Insights Across All Days

- SBI Mutual Fund is the clear AUM leader at ₹12.5 lakh crore by 2025 — nearly double its 2022 AUM of ₹6.05L Cr
- SIP inflows grew 2.7× from ₹11,517 Cr (Jan 2022) to ₹31,002 Cr (Dec 2025) — all-time high at year-end
- Investor folio count almost exactly doubled from 13.26 Cr to 26.12 Cr in 4 years — ~97% growth
- Banking sector dominates equity holdings at ₹62,840 Cr (19.3% of all equity fund market value)
- 26–35 age group is the largest investor segment (40.7%), and T30 cities contribute 2× the SIP value of B30 cities
- NAV return correlations near zero across all 40 funds — including between Regular and Direct plans of the same fund — confirming the NAV series are generated independently rather than reflecting shared market co-movement
- Small Cap funds carry the highest daily VaR (-3.5%) while Liquid funds approach near-zero risk — ordering matches risk labels in fund_master
- NAV growth is smooth at ~12-14%/year with no visible correction — the dataset does not exhibit the volatile bull-run/drawdown pattern of real markets
- 73.5% of equity funds show Low HHI sector concentration — broadly diversified portfolios
- The fund scorecard composite (0–100) produces a sensible spread of 12–86 across 40 funds, demonstrating a replicable, transparent ranking methodology

Important Data Characteristics

Several findings across this project point to the NAV time series in this dataset being generated independently per fund rather than driven by a shared underlying market:

- Even the Regular and Direct plans of the same fund (which hold identical portfolios) show near-zero daily return correlation — expected value in real markets is ~+0.99
- Alpha/Beta regressions against Nifty 100 have mean R^2 under 1% across all 40 funds
- Tracking error for the top-5 funds vs Nifty 50 exceeds 20% annualised — far above real-world levels
- NAV growth is uniformly smooth across all fund types, with no shared market events visible

These findings are reported honestly throughout the notebooks and dashboard rather than papered over. In a real-world dataset, the same analytics pipeline would produce fund correlations, Beta values, and rolling Sharpe patterns that reflect genuine market co-movement.

Git Commit History

- Day 1: Data ingestion complete
- Day 2: Cleaned data + SQLite DB loaded
- Fix sql/schema.sql (was overwritten with run_day2.py docstring) — Day 2 complete
- Day 3: EDA notebook complete — 15+ charts, 10 documented insights
- Day 4: Fund Performance Analytics complete — CAGR, Sharpe, Sortino, Alpha/Beta, drawdown, scorecard
- Day 5: Power BI dashboard complete — bluestock_mf_dashboard.pbix, PDF, 4 page PNGs
- Day 6: Advanced Analytics complete — VaR/CVaR, rolling Sharpe, cohort analysis, SIP continuity, recommender, HHI